 

# CAPE TOWN PORTNET, South Africa

WMO: **688170**

Lat: **33.9206S** Lon: **18.4431E** Elev: **2** StdP: **101.30** Time Zone: **2.00 (E02)** Period: **98-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **7** | **9.2** | **10.2** | **0.3** | **3.9** | **18.7** | **1.7** | **4.3** | **16.9** | **14.1** | **14.5** | **12.6** | **14.0** | **1.6** | **240** | **0.455** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **2** | **7.2** | **32.1** | **19.4** | **29.9** | **18.8** | **28.1** | **18.4** | **21.2** | **29.2** | **20.5** | **28.2** | **19.8** | **27.1** | **4.5** | **120** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **18.4** | **13.3** | **24.9** | **17.6** | **12.6** | **23.7** | **17.0** | **12.1** | **23.1** | **61.4** | **29.4** | **58.7** | **28.0** | **56.5** | **27.2** | **24.5** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **15.4** | **13.6** | **12.1** | DB | **7.6** | **36.1** | **0.9** | **2.5** | **6.9** | **37.9** | **6.3** | **39.3** | **5.8** | **40.7** | **5.1** | **42.5** | (4) |
| (5) | | | | WB | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | **Temperatures, Degree-Days and Degree-Hours** | DBAvg | | **18.3** | **22.1** | **22.2** | **20.8** | **19.0** | **16.7** | **15.0** | **14.9** | **14.9** | **16.1** | **18.2** | **19.2** | **21.1** | (6) |
| (7) | | DBStd | | **3.70** | **2.73** | **2.75** | **2.92** | **3.00** | **2.23** | **1.87** | **2.14** | **2.03** | **2.51** | **2.80** | **2.71** | **2.63** | (7) |
| (8) | | HDD10.0 | | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | (8) |
| (9) | | HDD18.3 | | **556** | **2** | **2** | **7** | **25** | **61** | **102** | **111** | **108** | **78** | **37** | **19** | **4** | (9) |
| (10) | | CDD10.0 | | **3041** | **376** | **343** | **334** | **271** | **209** | **151** | **151** | **153** | **182** | **253** | **276** | **343** | (10) |
| (11) | | CDD18.3 | | **554** | **119** | **111** | **83** | **46** | **12** | **2** | **3** | **3** | **10** | **32** | **45** | **89** | (11) |
| (12) | | CDH23.3 | | **2698** | **628** | **576** | **392** | **222** | **44** | **11** | **13** | **13** | **47** | **154** | **186** | **411** | (12) |
| (13) | | CDH26.7 | | **757** | **187** | **168** | **116** | **70** | **6** | **1** | **1** | **1** | **11** | **41** | **46** | **109** | (13) |
| (14) | **Wind** | WSAvg | | **4.4** | **5.9** | **5.3** | **4.7** | **3.9** | **2.9** | **3.1** | **3.3** | **3.4** | **4.3** | **4.9** | **5.6** | **5.5** | (14) |
| (15) | **Precipitation** | PrecAvg | | **574** | **19** | **16** | **22** | **44** | **77** | **109** | **82** | **87** | **46** | **33** | **22** | **19** | (15) |
| (16) | | PrecMax | | **829** | **75** | **67** | **86** | **172** | **198** | **248** | **158** | **230** | **112** | **135** | **80** | **54** | (16) |
| (17) | | PrecMin | | **349** | **0** | **0** | **0** | **6** | **14** | **23** | **0** | **21** | **4** | **0** | **0** | **0** | (17) |
| (18) | | PrecStd | | **119** | **19** | **18** | **24** | **31** | **48** | **57** | **41** | **50** | **29** | **29** | **21** | **15** | (18) |
| (19) | **Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures** | 0.4% | DB | | **33.9** | **34.2** | **33.7** | **32.5** | **27.4** | **25.0** | **25.1** | **24.9** | **27.8** | **31.4** | **30.9** | **32.9** | (19) |
| (20) | | | MCWB | | **20.8** | **20.1** | **19.7** | **18.4** | **15.6** | **14.3** | **14.2** | **15.0** | **15.5** | **18.0** | **17.1** | **21.0** | (20) |
| (21) | | 2% | DB | | **30.9** | **30.7** | **29.6** | **28.3** | **24.3** | **21.9** | **22.0** | **21.9** | **24.0** | **27.2** | **27.5** | **29.4** | (21) |
| (22) | | | MCWB | | **20.0** | **19.6** | **19.0** | **17.4** | **15.2** | **12.8** | **13.7** | **14.0** | **14.4** | **16.6** | **16.9** | **19.1** | (22) |
| (23) | | 5% | DB | | **28.5** | **28.5** | **27.2** | **25.5** | **22.1** | **19.8** | **19.9** | **19.7** | **21.4** | **24.7** | **25.2** | **27.3** | (23) |
| (24) | | | MCWB | | **19.3** | **18.9** | **18.5** | **16.6** | **15.0** | **12.9** | **13.4** | **13.4** | **13.8** | **16.1** | **16.3** | **18.5** | (24) |
| (25) | | 10% | DB | | **26.6** | **26.5** | **25.2** | **23.3** | **20.4** | **17.9** | **18.1** | **18.0** | **19.5** | **22.5** | **23.4** | **25.4** | (25) |
| (26) | | | MCWB | | **19.1** | **18.7** | **18.1** | **16.1** | **14.5** | **12.9** | **12.9** | **12.8** | **13.4** | **15.5** | **15.8** | **18.0** | (26) |
| (27) | **Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures** | 0.4% | WB | | **22.8** | **22.4** | **21.2** | **19.9** | **17.5** | **15.9** | **15.8** | **16.2** | **16.3** | **19.6** | **19.2** | **21.8** | (27) |
| (28) | | | MCDB | | **31.0** | **29.8** | **29.2** | **29.2** | **23.5** | **20.6** | **22.7** | **22.0** | **26.3** | **30.6** | **28.7** | **32.2** | (28) |
| (29) | | 2% | WB | | **21.3** | **20.8** | **20.2** | **18.7** | **16.5** | **14.8** | **14.9** | **15.2** | **15.3** | **18.2** | **18.0** | **20.4** | (29) |
| (30) | | | MCDB | | **29.1** | **28.0** | **27.3** | **27.7** | **21.6** | **18.2** | **19.6** | **21.0** | **22.1** | **26.5** | **25.3** | **28.9** | (30) |
| (31) | | 5% | WB | | **20.5** | **20.0** | **19.3** | **17.8** | **15.8** | **14.2** | **14.3** | **14.4** | **14.7** | **17.1** | **17.3** | **19.6** | (31) |
| (32) | | | MCDB | | **28.0** | **27.2** | **26.0** | **25.2** | **20.4** | **17.1** | **18.4** | **19.1** | **19.9** | **24.1** | **23.8** | **27.1** | (32) |
| (33) | | 10% | WB | | **19.7** | **19.3** | **18.5** | **17.1** | **15.2** | **13.7** | **13.7** | **13.6** | **14.1** | **16.3** | **16.7** | **18.9** | (33) |
| (34) | | | MCDB | | **26.7** | **26.1** | **24.5** | **23.1** | **19.1** | **16.5** | **17.3** | **17.3** | **18.3** | **22.3** | **22.9** | **25.5** | (34) |
| (35) | **Mean Daily Temperature Range** | | MDBR | | **7.1** | **7.2** | **7.0** | **6.6** | **5.9** | **5.5** | **5.8** | **5.6** | **5.4** | **6.2** | **6.1** | **6.6** | (35) |
| (36) | | 5% DB | MCDBR | | **10.9** | **10.9** | **10.8** | **11.2** | **9.9** | **9.8** | **9.8** | **9.4** | **10.0** | **10.8** | **9.9** | **10.5** | (36) |
| (37) | | | MCWBR | | **4.4** | **4.3** | **4.2** | **4.1** | **4.5** | **5.0** | **5.0** | **4.7** | **4.3** | **4.4** | **4.3** | **4.7** | (37) |
| (38) | | 5% WB | MCDBR | | **10.7** | **9.7** | **10.0** | **11.0** | **8.2** | **6.8** | **7.8** | **8.8** | **9.0** | **10.5** | **9.5** | **10.7** | (38) |
| (39) | | | MCWBR | | **4.6** | **4.2** | **4.1** | **4.4** | **4.2** | **4.0** | **4.3** | **4.6** | **4.1** | **4.3** | **4.3** | **4.7** | (39) |
| (40) | **Clear-Sky Solar Irradiance** | taub | | | **0.343** | **0.345** | **0.336** | **0.323** | **0.322** | **0.317** | **0.319** | **0.334** | **0.344** | **0.353** | **0.340** | **0.344** | (40) |
| (41) | | taud | | | **2.540** | **2.543** | **2.565** | **2.593** | **2.589** | **2.576** | **2.566** | **2.521** | **2.456** | **2.446** | **2.526** | **2.528** | (41) |
| (42) | | Ebn at Noon | | | **994** | **972** | **945** | **902** | **847** | **824** | **842** | **875** | **920** | **951** | **991** | **996** | (42) |
| (43) | | Edh at Noon | | | **109** | **105** | **96** | **83** | **74** | **70** | **75** | **88** | **105** | **115** | **110** | **112** | (43) |
| (44) | **All-Sky Solar Radiation** | RadAvg | | | **8.21** | **7.33** | **5.88** | **4.30** | **2.97** | **2.50** | **2.79** | **3.53** | **4.88** | **6.45** | **7.66** | **8.30** | (44) |
| (45) | | RadStd | | | **0.18** | **0.21** | **0.22** | **0.21** | **0.19** | **0.14** | **0.17** | **0.17** | **0.20** | **0.30** | **0.22** | **0.19** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (12 neighbors) | **+0.36** | **+0.49** | **N/A** | **N/A** | **+0.53** | **+0.47** | **-2** | **-74** | **+111** | **+61** | (47) |

Nomenclature: See separate page